



Lexical competition in the process of Cantonese tone merging: Diverse Impact Mechanisms Across Different Individuals and Tone Pairs

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Abstract

Lexical competition has been shown to exert an inhibitory influence on phonemic mergers in phonetic evolution. This study investigates the merging process of Cantonese tone pairs, examining the lexical competition effect from an individual perspective. Results reveal that lexical competition influences tone production even in speakers without clear mergers, helping maintain tonal contrasts while subtly altering pitch distributions, which may signal future change. For participants merging only one tone pair, the effect of lexical competition varies: it consistently inhibits merging for the T4 (21) – T6 (22) pair, but for the T3 (33) – T6 (22) pair, its influence manifests in three distinct patterns—no effect, inhibition, or promotion of merging. Conversely, in participants merging all three tones, lexical competition shows minimal impact. This research elucidates the diverse mechanisms through which lexical competition shapes tone-merging processes.

Index Terms: tone mergers, tone production, lexical competition, individual difference, Cantonese

1. Introduction

A key question in historical linguistics is why some sounds are more prone to change while others remain stable over time [1]. Research has shown that internal linguistic factors, such as word frequency, and lexical competition (e.g., minimal pairs, neighborhood density) are crucial in promoting or inhibiting phonetic changes [2, 3, 4, 5, 6].

Lexical competition plays a crucial role in phonetic evolution. In French, for example, minimal pair nasal vowel competitors reduce nasal coarticulation in Consonant-Vowel-Nasal words, suggesting that lexical competition can inhibit phonetic changes by enhancing phonemic distinctiveness [3]. Similarly, studies on natural spoken English found that phonological contrasts between lexical items affect phonetic change, such as in word-initial plosives, where minimal pairs with voiced or voiceless plosives alter Voice Onset Time (VOT) [4][5]. A corpus-based research confirms that phonemes distinguishing fewer minimal pairs are more likely to merge, while those distinguishing many minimal pairs are less likely to merge [6]. In Bardi, a language in northern Western Australia, phonemes with fewer minimal pairs also tend to merge [1]. Generally, lexical competitors lead speakers to increase phonetic distinctiveness through strategies like

altering duration, VOT, and other articulatory features. A study proposed a model explaining why high neighborhood density (the presence of many competitors) aids spoken word production, suggesting that lexical activation occurs at two levels: lexical and segmental. Feedback from the segmental level (i.e., similar words) boosts target word activation, making it easier for the speaker to produce it [7]. However, most of the above research has focused on the sound change of vowels and consonants in non-tonal languages.

Recent studies have examined tonal change in Chinese dialects, including Cantonese [8, 9], Hakka [10], and the checked tones in Jin [11], focusing on production, perception, and structural factors. However, they largely overlook lexical competition, which may influence the maintenance of tonal contrasts. One recent study on Cantonese provides preliminary evidence that lexical competition helps preserve tonal distinctions by affecting listeners' acceptance of phonetic variants [12]. Yet, this study only contrasted merged versus non-merged tone pairs, without accounting for whether individual participants actually exhibited tonal mergers. Moreover, the effect of lexical competition on production—not just perception—remains underexplored. In addition, most prior research has focused on group-level patterns, often neglecting the role of individual variation in language change [13]. Our previous work on Cantonese tonal mergers identified five distinct merging patterns among younger speakers [14], but did not investigate the cognitive or lexical factors that might explain these individual differences.

We hypothesize that lexical competition, known to influence mergers in non-tonal languages, also affects tonal mergers with varying effects across individuals. The diverse merger patterns in Guangzhou Cantonese offer an ideal setting to study this. This paper will further explore the phenomenon of tonal mergers between T3 (represented as 33 in the five-degree tonal system), T4 (21), and T6 (22) in Guangzhou Cantonese, examining how lexical competition factors affect individuals with different tonal merger patterns.

2. Methods

2.1. Production task

2.1.1. Participants and Materials

Fifty-three native Cantonese speakers (27 male, 26 female; age 18 – 30, M = 23.6, SD = 3.32) from several universities

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participated. All participants were native Cantonese speakers who used Cantonese in daily life and did not speak any other Chinese dialects. They had learned Mandarin and English in educational contexts. None had musical training or speech/hearing impairments. The production task lasted 20 minutes, and participants received compensation. We selected 19 pairs of syllables that simultaneously contained T4, T3, and T6 (Competitive), 19 pairs of syllables with minimal pairs only at T4 and T6 (Non-competitive_1), and 19 pairs of syllables with minimal pairs only at T3 and T6 (Non-competitive_2). In total, 133 example characters were chosen. The selected characters were semantically clear and monophonic, avoiding any ambiguity in pronunciation. Table 1 provides examples of the characters used.

Table 1: Examples of recorded words (competitive vs. non-competitive)

Competitive			
Syllable	T3_A	T4_A	T6_A
jiu	婁(want)	遙(far)	耀(shine)
Non-competitive 1			
Syllable	T3	T4_B	T6_B
won	null	皇(king)	旺(swam)
Non-competitive 2			
Syllable	T3_C	T4	T6_C
pai	閉(close)	null	幣(coin)

2.1.2. Procedures

The recordings were made in quiet, soundproof rooms using xRecorder software [15]. The target presentation was randomized. The sampling rate was 22.05 kHz, with 16-bit depth and a mono-channel configuration.

2.2. Analysis

2.2.1. Normalization

Production data were annotated and 11-point F0 contours were extracted in Praat [16] by the first author, a native Cantonese speaker. The non-contrastive T4–T6 pair “来 (‘come’) – 睐 (‘look favorably upon’)” was excluded due to participants’ confusion over the tone of “睐.” Four participants were also removed due to poor recording quality, leaving data from 49 participants. Of 6,419 word tokens, 5,914 were retained (9.21% loss due to poor audio, misreadings, or creaky voice), with 71.49% corresponding to T4. The first and last sampling points were removed, and F0 values were normalized using the lz-score method to reduce inter-speaker and gender-related variability [17].

$$z_i' = \frac{y_i - m_y}{s_y} \quad (1)$$

Where y_i represents the logarithmic value of the tone’s F0; m_y is the mean of the logarithmic F0 values for all tones of the speaker; s_y is the standard deviation of the logarithmic F0 values for all tones of the speaker.

2.2.2. Random Forest

To classify tone pairs, we adopted the Random Forest model [18], an established method for tone classification [19]. Eight participants without tone merger in perception ($\geq 80\%$ accuracy in the perception experiment) or production (auditorily judged by the first author) were selected as “correct pronunciation” samples. Another eight participants with perceptually identifiable mergers (T3–T6 or T4–T6) ensured

balanced group sizes. Acoustic features (average pitch, slope, curvature) were extracted for three tone pairs. The Random Forest classifier, implemented in scikit-learn [20], was used to detect merger patterns, with the number of trees ($n_estimators$) optimized via 5-fold cross-validation. Model outputs were cross-validated against perceptual labels and pitch contour divergence before further analysis. The best model then predicted pronunciation categories. Given the potential effects of lexical competition, models were trained and evaluated separately under competitive and non-competitive conditions. Optimal tree numbers were: competitive – 20 (T3–T4, T3–T6), 40 (T4–T6); non-competitive – 80 (T4–T6), 140 (T3–T6).

3. Results

3.1. Overall Results

Using predictions from the Random Forest model, we calculated the overall production accuracy for each tone pair in both competitive and non-competitive states. Figure 1 presents boxplots of production accuracy across states. The median accuracy for the T4–T6 and T3–T6 tone pairs was around 75%, with a slightly lower accuracy for T3–T6 in the non-competitive state. The highest accuracy, around 90%, was observed for the T3–T4 tone pair.

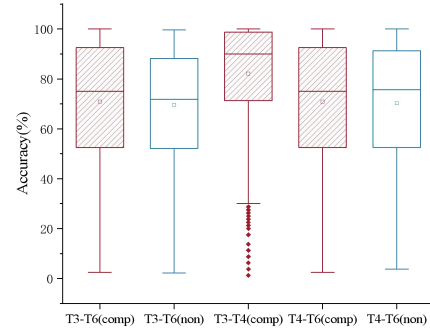


Figure 1: Boxplots of overall production accuracy for each tone pair (competitive vs non-competitive).

To investigate the effects of gender, competition, age, and their interactions on production accuracy, this study employed a linear mixed-effects model using the lmer() function from the lme4 package in R (version 4.3.0) [21, 22]. The model formula was:

$$\text{Accuracy} \sim \text{Gender} + \text{Competition} + \text{Age} + \text{Tone Pair} + \text{Gender} * \text{Age} + \text{Tone_Pair} * \text{Competition} + (1 | \text{Participant})$$

The results indicated that competition state ($\beta = 2.054e-02$, $p < 0.05$) and participant age ($\beta = 1.514e-02$, $p < 0.05$) significantly affected production accuracy. The interaction between tone pair and competition state also significantly impacts ($\beta = -3.917e-02$, $p < 0.01$). No significant effects were found for other variables ($p > 0.05$).

3.2. Results of Grouping

3.2.1. Accuracy

Production accuracy above 80% was defined as no tone merger, and below 80% as merger. Accuracy predicted by the Random Forest model across five conditions (competitive and non-competitive for three tone pairs) classified participants into six groups, each representing distinct merger patterns under varying competition contexts. These six patterns can be

further categorized into four directions: **A:** G1, which maintains the three-tone pattern. **B:** G2 and G3, where the merger tendency between T3 and T6 is more pronounced. **C:** G4 and G5, where the merger tendency between T4 and T6 is more pronounced. **D:** G6, where all three tones merge. For each group, the production patterns in both competitive and non-competitive states remained consistent across participants.

Table 2 presents the random forest model predictions of the average production accuracy for each tone pair in competitive and non-competitive states for each group, with the number of participants listed for each group.

Table 2: Average production accuracy for each tone pair (competitive vs. non-competitive states). Gray shading indicates accuracy below 80%.

Group	number	Competitive			Non-competitive	
		T3-T4	T3-T6	T4-T6	T3-T6	T4-T6
G1	8	97.03%	90.07%	93.60%	91.82%	91.92%
G2	15	91.75%	71.36%	85.64%	72.89%	83.73%
G3	7	91.01%	66.51%	84.03%	63.30%	73.27%
G4	5	84.41%	67.67%	69.82%	70.19%	67.27%
G5	11	87.39%	73.37%	56.88%	82.18%	53.12%
G6	3	65.15%	57.48%	56.42%	58.03%	59.14%

To examine the effects of gender, competitive state, age, and their interactions on production accuracy, we fitted a Linear Mixed Effects (LME) model to test the data for the T4-T6 and T3-T6 tone pairs across Groups G2 to G6. The model formula is as follows:

$$\text{Accuracy} \sim \text{Gender} + \text{Competition} + \text{Age} + \text{Gender} * \text{Age} + (1|\text{Participant})$$

No significant main effects of gender, age, or their interaction were found for any group or tone pair ($p > 0.05$). Lexical competition significantly affected production accuracy for the T4-T6 pair in G3 ($\beta = 0.1069$, $p < 0.001$) and the T3-T6 pair in G5 ($\beta = -0.0880$, $p < 0.001$), as highlighted in bold in Table 2. No significant competition effects were observed for other groups and tone pairs ($p > 0.05$).

3.2.2. Tone Merger Patterns

After classifying the groups, we plotted the F0 curves for each tone in Groups G1-G6 under both competitive and non-competitive states (Figure 2).

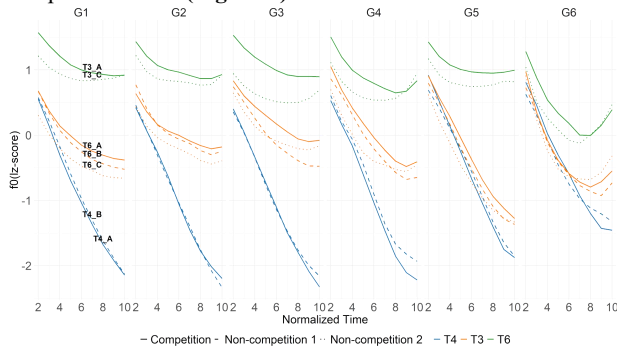


Figure 2: Pitch Contour Plots of the Six Merging Patterns.

The solid lines represent the F0 curves for the three tones in the competitive state, while the dashed lines represent the F0 curves for the three tones in the non-competitive state. “Non-competition 1” refers to the condition when the lexical competitor T3 is absent, and “Non-competition 2” refers to the

condition when the lexical competitor T4 is absent. For ease of reference, we label the F0 curves for the three tones under lexical competition as T4_A, T3_A, and T6_A. When the lexical competitor T3 is absent, T4 is labeled as T4_B and T6 as T6_B. When the lexical competitor T4 is absent, T6 is labeled as T6_C and T3 as T3_C. Although the labels are only indicated for Group G1 in Figure 2, they apply to all groups.

To examine the effects of competitive state on F0 production under different merger patterns, Growth Curve Analysis (GCA) [23] was conducted using linear mixed-effects models fitted with the lme4 package [21] in R (version 4.3.0) [22]. GCA models time-varying acoustic data by including polynomial terms of time as predictors. The model formula is as follows:

$$F0 \sim \text{ot1} + \text{ot2} + \text{Gender} + \text{Age} + \text{Competition} + \text{ot1:Competition} + \text{ot2:Competition} + (1|\text{Participant})$$

In GCA, ot1 corresponds to the first-order polynomial term, representing the slope of the F0 contour, which reflects the pitch’s rising or falling trend. ot2 corresponds to the second-order polynomial term, representing the curvature of the contour, indicating its concavity or convexity.

Table 3: p-values of GCA Models in six groups.

	G1	G2	G3	G4	G5	G6
Comp	$p < 0.001$	$p < 0.001$	$p < 0.001$	$p < 0.001$	$p < 0.001$	$p < 0.001$
T3 ot1:Comp	$p < 0.001$	n.s.	$p < 0.05$	$p < 0.001$	n.s.	$p < 0.001$
ot2:Comp	n.s.	n.s.	n.s.	n.s.	n.s.	n.s.
Comp	n.s.	n.s.	n.s.	$p < 0.05$	n.s.	n.s.
T4 ot1:Comp	n.s.	n.s.	n.s.	n.s.	n.s.	n.s.
ot2:Comp	n.s.	n.s.	n.s.	n.s.	n.s.	n.s.
Comp	$p < 0.001$	n.s.	$p < 0.001$	$p < 0.001$	$p < 0.01$	n.s.
T6_B ot1:Comp	n.s.	n.s.	$p < 0.05$	n.s.	n.s.	n.s.
ot2:Comp	n.s.	n.s.	n.s.	n.s.	n.s.	n.s.
Comp	$p < 0.001$	$p < 0.001$	$p < 0.001$	$p < 0.001$	$p < 0.001$	n.s.
T6_C ot1:Comp	n.s.	n.s.	n.s.	$p < 0.001$	$p < 0.05$	$p < 0.05$
ot2:Comp	n.s.	n.s.	n.s.	n.s.	n.s.	n.s.

We combine the tone pattern distributions from the pitch contour graphs and the results of the GCA analysis to examine the tonal production patterns under competitive and non-competitive conditions across the groups.

In G1, T6_B shows a significant difference in average pitch ($p < 0.001$), while T4 shows no change. T6_C and T3 also show significant differences in average pitch ($p < 0.001$), with T3 additionally differing in slope ($p < 0.001$). In G2, T4 and T6_B show no significant differences in slope, average pitch, or curvature ($p > 0.05$), but T6_C and T3 show significant differences in average pitch ($p < 0.001$). In G3, T6_B differs significantly in average pitch and slope ($p < 0.001$, $p < 0.05$), while T4 remains stable ($p > 0.05$). T6_C and T3 also differ significantly in average pitch ($p < 0.001$), with T3 differing in slope ($p < 0.05$). In G4, T4 shows a significant difference in average pitch ($p < 0.05$), while T6_B, T6_C, and T3 show significant differences in average pitch ($p < 0.001$), and T6_C, and T3 show significant differences in slope ($p < 0.001$). In G5, T6_B and T3 show significant differences in average pitch ($p < 0.01$, $p < 0.001$), while T4 remains stable ($p > 0.05$). T6_C shows significant differences in both average pitch and slope ($p < 0.05$). In G6, T4 and T6_B show no significant differences ($p > 0.05$). T6_C differs significantly in slope, and T3 differs significantly in both average pitch and slope ($p < 0.001$).

4. Discussion

This study investigates how lexical competition influences tone merging across individuals, focusing on six patterns. Although competition has an overall effect on production accuracy ($p < 0.05$), its influence differs across tone pairs, reflecting both common and divergent trends.

We first provide an overview of the common patterns in tone distribution under different conditions. Two key trends were identified across the groups: **1)** T4 remains unaffected by competition and stays stable at the bottom, with no significant differences in average pitch, slope, or curvature ($p > 0.05$), except for G4; **2)** Lexical competition increases the average pitch of T3 and T6 compared to the non-competitive condition ($p < 0.01$, except for G2). This pattern holds both in the group without tone mergers (G1) and in groups where tone mergers occur (G2 to G5).

The stability of T4 is due to its falling tone characteristics, requiring less articulatory effort [24]. Moreover, it remains stable since T4 is positioned at the bottom of the Cantonese tone system. In competitive conditions, the average pitch of T3 and T6 increases, expanding the acoustic space between these tones and aligning with the communicative clarity principle. Previous studies suggest speakers adjust pronunciation to enhance word recognition, particularly for phonetically similar words, by increasing dispersion for better listener distinction [25][26]. This explains why T3 and T6 have higher average pitch in competition.

Next, we will analyze the effect of lexical competition on different tone pairs and merging patterns. Starting with the T4-T6 tone pair, under the premise that T4 stays at the bottom, if the competitor T3 is absent, the decreased height of T6 compresses the acoustic space between T4 and T6, encouraging a merger. This parallels vowel dispersion in easily recognized words (fewer competitors) versus hard-to-recognize words (more competitors), where harder words show greater F1 and F2 spread [26]. The presence of lexical competitors T3 prevents the merger of the T4 and T6 tone pair. This finding is consistent with previous studies on phonological evolution in English and Bardi languages, which also show that lexical competition can inhibit sound mergers.

We examine how lexical competition influences tone merging in the T3-T6 pair by breaking down the consistent patterns observed in G1 to G5 into three combinations. We focus on shared characteristics, with group-specific details discussed later.

1)T6_B, T6_C & T3_A: In this case, the average height of T6_B and T6_C decreases under non-competitive conditions, and the acoustic space between the T3-T6 tone pair increases compared to the competitive state (T3_A & T6_A), which inhibits further merging of the T3-T6 tone pair. Thus, the non-lexical competitive state suppresses the merger of the T3-T6 tone pair.

2)T6_C & T3_C: Although the average height of T6_C decreases, the decrease in T3_C is similar, and the acoustic space remains relatively consistent between lexical and non-lexical competitive conditions (T3_A & T6_A). Given the relative nature of tone height, lexical competition has little impact on this distribution.

3)T6_B & T3_C: Compared to the competitive state (T3_A & T6_A), the acoustic space in this case is smaller, and the presence of lexical competitors inhibits further merging.

Compared to lexical competition's clear inhibition of the T4-T6 merger, its effect on T3-T6 varies, likely due to tone distribution. T3 and T6 occupy mid-low pitch regions with

greater variability than T4 [8]. Lexical competition inhibits mergers depending on tone distribution: T4's stability blocks T4-T6 merging, while competition's effect on T3-T6 varies (promotion, inhibition, or none). Previous studies suggest that vowel F1 and F2 distributions may be influenced by "recognizability" [26], though others argue that consonantal environments, not recognition ease, drive these patterns [27]. Similarly, the impact of lexical competitors on the T3-T6 tone pair requires further exploration, considering contextual factors like vowel and consonant distributions.

Finally, we focus on the peculiarities of G3, G5, and G6. G3 shows lower accuracy for the T3-T6 pair and a stronger merging trend than G2. Under non-competitive conditions, T3_C and T6_C decrease inconsistently, with T3_C showing a greater reduction, narrowing the acoustic gap between T3 and T6. The T4-T6 pair in G3 also shows a significant merging trend ($84.03\% \rightarrow 73.27\%$, $p < 0.05$), suggesting a greater likelihood of further merging compared to other groups. Groups with deeper mergers may exhibit more tone system variability, potentially linked to tone evolution shifts [28].

G5 shows lower accuracy and a stronger merging trend for the T4-T6 pair than G4. The T3-T6 accuracy significantly differs between competitive and non-competitive conditions ($p < 0.001$), shifting from merged (73.37%) to non-merged (82.18%). However, the tone pattern distribution (Figure 2) shows no actual merger in the T3-T6 pair for G5. The T3-T6 distribution in G5 is skewed compared to G1, indicating that the random forest algorithm reflects relative differences, not absolute merging. This may be due to the features used (e.g., height, slope, curvature), which might not fully capture tonal variations. Future studies should improve methods for assessing tonal evolution.

G6 shows the most pronounced merging, with a significant reduction in the acoustic space between the three tones. The distribution patterns under competitive and non-competitive conditions are less regular than in other groups, and lexical competition has minimal impact on tone merging.

This research extends the role of lexical competition to tone production, showing its complex influence varies by tone pair, and merging pattern. The consistent inhibition of merging in T4-T6 versus the varied effects in T3-T6 highlight how lexical competition differentially shapes the direction and dynamics of tone evolution depending on the specific merger pattern.

5. Conclusions

In conclusion, this study examined how lexical competition influences tone mergers among T3, T4, and T6 in Cantonese using a production task. Lexical competition was found to reshape tone distribution even in speakers without clear merging, helping maintain distinctions across tone pairs, though sometimes altering pitch distribution in ways that could lead to future change. Among merging speakers, competition consistently inhibited further T4-T6 merger, while its effect on T3-T6 varied with tone distribution. In cases of full three-way mergers, lexical competition had minimal influence. These results underscore the interaction between lexical competition and tone system dynamics. Future research should further investigate how competition, phonetic context, and segmental factors jointly contribute to the trajectory of tone merger, tone evolution and refine methods for quantifying tone merger degrees.

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